Algorithm 2 Kinematics of the parallel continuum robot computation $f(\boldsymbol{u}_a) = \boldsymbol{y}_e$

Initialize $\boldsymbol{q}^0 = [\boldsymbol{q}_1^0;\ \boldsymbol{q}_2^0;\ \boldsymbol{q}_3^0;\ \boldsymbol{q}_4^0]$
for $i = 1, \ldots, \text{maxiter}$ **do**
  Form the residual: ▷ simulate states and gradients

$$\begin{cases} \boldsymbol{b}_1 \leftarrow \boldsymbol{F}_1(\boldsymbol{q}_1^{i-1}) + \boldsymbol{M}_1 \boldsymbol{g} \\ \qquad\qquad \vdots \\ \boldsymbol{b}_4 \leftarrow \boldsymbol{F}_4(\boldsymbol{q}_4^{i-1}) + \boldsymbol{M}_4 \boldsymbol{g} \end{cases}$$

  Compute $\boldsymbol{A}_j(\boldsymbol{q}_j^{i-1})$ and $\boldsymbol{H}_j$ $(j \in [1, 4])$ to form $\boldsymbol{A}$, and $\boldsymbol{H}_a, \delta_a(\boldsymbol{q}^{i-1})$
  Find $d\boldsymbol{q}$: ▷ QP solve

$$\begin{cases} \boldsymbol{A}\, d\boldsymbol{q} = \boldsymbol{b} + \boldsymbol{H}_e^{\top} \boldsymbol{\lambda}_e + \boldsymbol{H}_a^{T} \boldsymbol{\lambda}_a \\ \boldsymbol{\delta}_a(\boldsymbol{q}^{i-1}) + \boldsymbol{H}_a\, d\boldsymbol{q} = \boldsymbol{u}_a \end{cases}$$

  $\boldsymbol{q}^i \leftarrow \boldsymbol{q}^{i-1} + \alpha d\boldsymbol{q}$ ▷ take solver step
  if $\|\boldsymbol{q}^i - \boldsymbol{q}^{i-1}\| < \epsilon$ **then**
    return $\boldsymbol{y}_e = \boldsymbol{\delta}_e(\boldsymbol{q}^{i-1})$ ▷ solver converged
  end if
end for